

Arsak Architectural Design System

A complete reference specification for building tactile, physical-depth UI components: 4-layer physical cards, micro-interaction buttons, ambient teal mesh glow, and high-performance WebGL shaders.

01 Design Philosophy: Tactile Physical Surface Engineering

Most modern websites look flat and generic because they rely on simple 1px gray borders and blurry drop shadows. This design system treats the digital screen as a **multi-layered physical glass and ceramic surface**. Every card, button, and container uses 4 distinct visual depth cues:

Physical Layer	Optical Technique	Perceived Real-World Effect
1. Specular Top Glaze	2px gradient white-to-transparent line	Direct sunlight catching the sharp top edge bevel of glass or acrylic.
2. Recessed Compartment	Subtle inset shadow + cool tint	Physical milled tray holding the category badge and icon.
3. Dual-Stage Depth Shadow	Short ambient shadow + long colored glow	Object elevated 8mm to 16mm off the desk with colored ambient bounce.
4. Bottom Extrusion Shelf	3.5px beveled shelf foot	Architectural weight so elements feel grounded rather than paper-thin.

02 Core Color Palette & Token Harmony

Daylight Canvas #EDF8F9 (Base)	Recessed Shelf #E8F6F8 (Headers)	Electric Lime #BCEB28 (Primary CTA)	Teal / Mint Flow #6bf5ff + #94ffd1
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Color Pairing Rule: Electric Lime (#BCEB28) is strictly reserved for *high-conversion triggers* (primary action buttons and urgent tags). The background and structural borders bathe in soft, refreshing *Teal / Mint* (#6bf5ff / #EDF8F9). This creates maximum contrast without visual fatigue.

03 The 4-Layer Interactive Card Architecture

Below is an exact visual rendering of the physical card anatomy used sitewide:

Architecture Module

LAYER.01

Dual-Compartment Physical Depth

Notice the specular white line at the very top, the recessed cool-tinted header tray, and the beveled shelf edge at the bottom.

• 100% GPU Accelerated

Hover Lift: -8px

```
/* Tailwind CSS Recipe for the 4-Layer Interactive Card */
<div className="relative group rounded-2xl overflow-hidden bg-white
  border border-[rgba(107,245,255,0.25)]
  shadow-[0_14px_34px_-8px_rgba(107,245,255,0.12),0_4px_12px_rgba(22,20,14,0.04)]
  hover:-translate-y-2 hover:shadow-[0_30px_65px_-12px_rgba(107,245,255,0.28)]
  hover:border-[rgba(107,245,255,0.45)] transition-all duration-300 transform-gpu">

  <!-- Layer 1: Top Specular Glaze Sheen -->
  <div className="pointer-events-none absolute inset-x-0 top-0 h-[2px]
    bg-gradient-to-r from-transparent via-white/95 to-transparent z-20" />

  <!-- Layer 2: Bottom 3D Extrusion Shelf -->
  <div className="pointer-events-none absolute inset-x-0 bottom-0 h-[3.5px]
    bg-gradient-to-r from-[#DCEAE9] via-[#BFE1E3] to-[#DCEAE9]
    border-t border-[#A8D5D8] z-20 shadow-[inset_0_1px_0_rgba(255,255,255,0.6)]" />

  <!-- Layer 3: Top Recessed Compartment Header -->
  <div className="p-4 bg-[#E8F6F8] border-b border-[rgba(107,245,255,0.20)]
    shadow-[inset_0_2px_4px_rgba(107,245,255,0.08)] flex justify-between">... </div>

  <!-- Layer 4: Clean Content Body -->
  <div className="p-5 bg-white">... </div>
</div>
```

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```
/* 1. Primary Electric Lime Button (.btn-lime) */
.btn-lime {
  background-color: #BCEB28;
  color: #001204; /* Deep Obsidian Ink */
  font-weight: 700;
  border-radius: 9999px;
  border: 1px solid rgba(13, 18, 4, 0.15);
  box-shadow: 0 4px 14px -2px rgba(188, 235, 40, 0.45),
              0 2px 4px rgba(13, 18, 4, 0.05),
              inset 0 1px 0 rgba(255, 255, 255, 0.6);
  transition: all 0.22s cubic-bezier(0.22, 1, 0.36, 1);
}
.btn-lime:hover {
  background-color: #AAdB1e;
  transform: translateY(-2px);
  box-shadow: 0 10px 24px -4px rgba(188, 235, 40, 0.6), inset 0 1px 0 #FFF;
}
.btn-lime:active {
  transform: scale(0.97); /* Physical tactile compression */
}
```

05 Atmospheric Background & 10 FPS WebGL Engine

Heavy 3D backgrounds often make sites sluggish and battery-draining. We solve this with a **two-tier performance strategy**:

Tier 1: CSS Architectural Dot Grid

Zero-CPU dot pattern rendered natively via CSS radial gradients:

```
background-image: radial-gradient(#A8D8E8 1.1px, transparent 1.1px);
background-size: 28px 28px;
```

Tier 2: 10 FPS Capped Shader Loop

Caps requestAnimationFrame to ~100ms ticks, reducing GPU consumption by 83%:

```
const TARGET_FPS = 10;
const FRAME_MS = 1000 / TARGET_FPS;
if (delta < FRAME_MS) return; // skips frames
```

```
// The Tri-Color WaterPlane Shader formula:
vec3 col = mix(uColor1, uColor2, smoothstep(0.0, 0.55, wave)); // #94ffd1 → #6bf5ff
col      = mix(col,    uColor3, smoothstep(0.45, 1.0,  wave)); // → #ffffff
col      = col * 1.2; // 1.2 Brightness lift for glowing daylight feel
```

06 Typography & Layout Standard Cheat Sheet

Role	Font Family	Tracking / Leading	Typical Tailwind Class
Display Headings	Space Grotesk (Bold 700)	-0.035em / 1.08 - 1.15	text-3xl lg:text-5xl font-bold tracking-tight
Section Subtitles	Plus Jakarta Sans (500)	Normal / 1.55 - 1.65	text-sm md:text-base text-[#4A453A] leading-relaxed
Status & Metrics	JetBrains Mono / font-mono	+0.04em uppercase	text-xs font-mono font-bold tracking-wider

Ready to drop into any new project: Copy the .btn-lime, .btn-secondary, and card layout classes into your index.css or Tailwind config. Use #EDF8F9 as your base canvas and add #BCEB28 for instant high-contrast visual hierarchy!